

GROUP PROJECT PROPOSAL

Your Group Number: 3

Section: 2

Your Group's Team Members: Kai Fan, Dawn Grant, Quang Huynh

Project Title: Car Prices Model Predictor

Dataset Name (If decided): <https://www.kaggle.com/code/khotijahs1/cars-price-prediction>

What is the problem statement?

Buying or selling a used car is difficult since prices vary widely depending on factors such as brand, model year, mileage, fuel type, color, condition, etc. This project aims to build a machine learning model that accurately predicts the selling price of a car using historical car listing data

Proposed Methodology (Discuss types of models, validation, evaluation plans):

Data preprocessing: handle missing values, remove duplicates/outliers, encode categorical variables (one-hot encoding/label encoding), feature scaling for numerical variables

EDA: correlation heatmaps (maybe), price distribution, feature importance visualization

Models: linear regression, decision tree regressor, random forest regressor

validation strategy: train/test split (80/20), cross-validation

evaluation metrics: MAE, MSE, RMSE, R^2 score

GitHub Link (Make sure the repository is public):

<https://github.com/quangshuynh/CSCI335-Project>